

Electrical ELHV Class 08/23/2022 classmate
22-23 Ev Storage 08/25/2022 Data _____ Page _____

48V, 15kWh

$$\text{Battery Ah} = C / \text{Voltage} \\ = 15000 / 48 = 300 \text{ Ah}$$

State of Charge (SoC) of battery is a measure of percentage of battery charged.

SoC of 0% means discharged battery

SoC of 100% means fully charged battery

We never let battery charged 100% charged or never let battery discharged to 0%.

It reduces life of battery.

O/P of battery varies with its SoC

For 48V LiON battery

Voltage varies from 43V to 56V depending up SoC.

↳ 0 SoC ↳ 100 SoC.

C rate of battery.

1C Charge/discharge. (0% to 100%) in 1 hour at power 15kW.

2C in 30 min, 30kW (you may not want to

0.1C means 10 hours, 1.5kW charge to 100%.

& it may not be at 0%.

Capacity of battery : - Aging with time.

Charge-Discharge cycles: - Every time charge or discharged.

When battery capacity goes down

O/P of battery goes down (from 15kWh to 12kWh)

at 80%.

(70%)(10.5kWh)

Usable energy each charge-discharge cycle is typically $x\%$. (may be 85%) of total capacity. (Depend on battery)

Depth of Discharge - For long life, never fully discharge or empty the battery (leave certain charge)

Also battery capacity reduces with each charge-discharge cycle

When battery capacity remaining or SoH is $y\%$ (typically 80%) of initial capacity the range gets proportionally reduced. battery life for EV is over & needs replacements.

at end of its life cycle

$$\begin{aligned}\text{Usable Capacity} &= x * y * C \\ &= 0.8 * 0.85 * C \\ &= 0.68 C\end{aligned}$$

So if in beginning it gives 100 km it will give you 68 km at its end of life cycle.

After end of life of 1st life, it can be used in UPS etc for 2nd life.

Battery pack made using cells

Life of battery is dependent on life of battery cells.

If 1 cell goes lower in charge the whole battery pack goes lower.

Cycle-life of battery cell is fundamental parameter, depending upon its chemistry, but it also depends on.

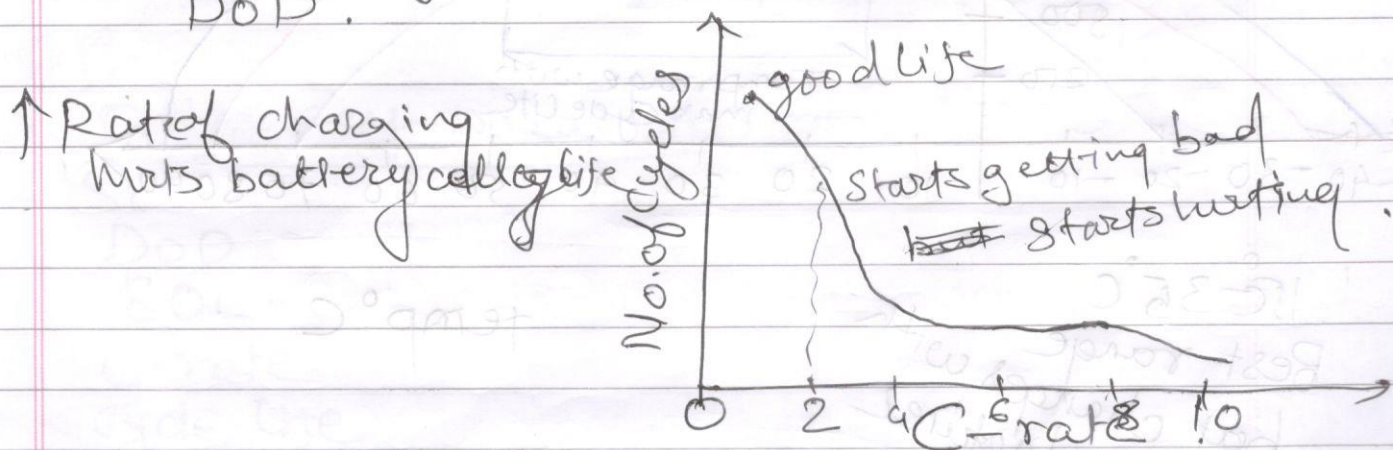
- C-rate of charging - discharging
- Temperature of its charging - discharging

Best for Li-ON battery.

- When used at 25°C or temp. of usage is 25°C
- When C-rates is less than 0.1C .
- When charging - discharging SoC is betⁿ 10% - 80% .

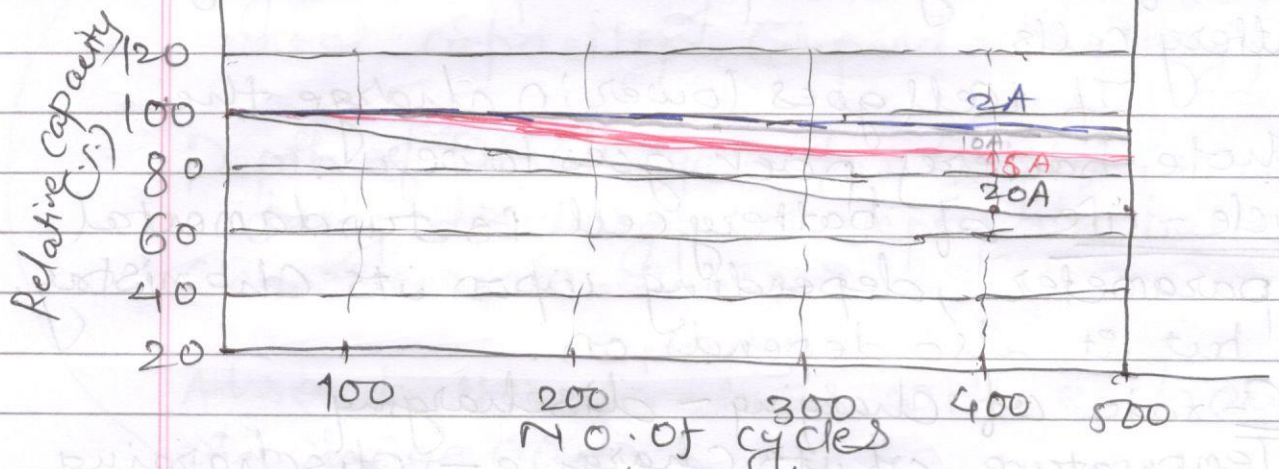
A Li-ON Cell manufacturer guarantees 1000 cycles under conditions like

- 1000 cycles when charged at $\text{C}/2$, discharged at C rate at 25°C with 85% DoD.

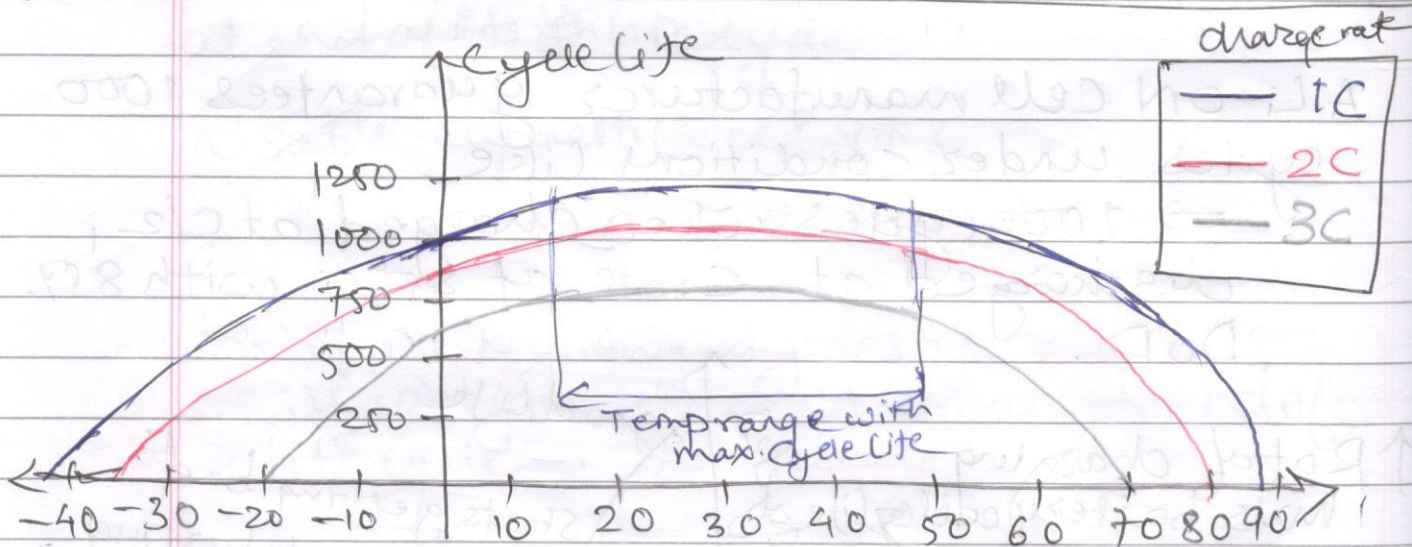


23 °C cycle life & discharge rate.

Charged at 2A to 4.2 V / cell
discharge at 2.5V & 20u / cell



When specified at C/2 rate & if charged at 3C rate, battery life may get reduced by factor of 5 or more.
(less than 400 cycles at 20 A, at 2A it can do around 1000, 1500 cycles)

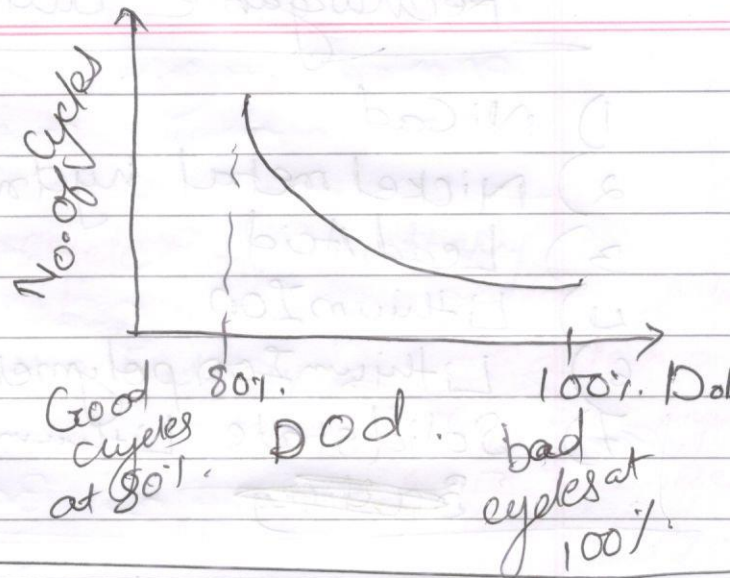


15°-35° C

Best range
but changes with
manufacture

temp °C

Depth of discharge

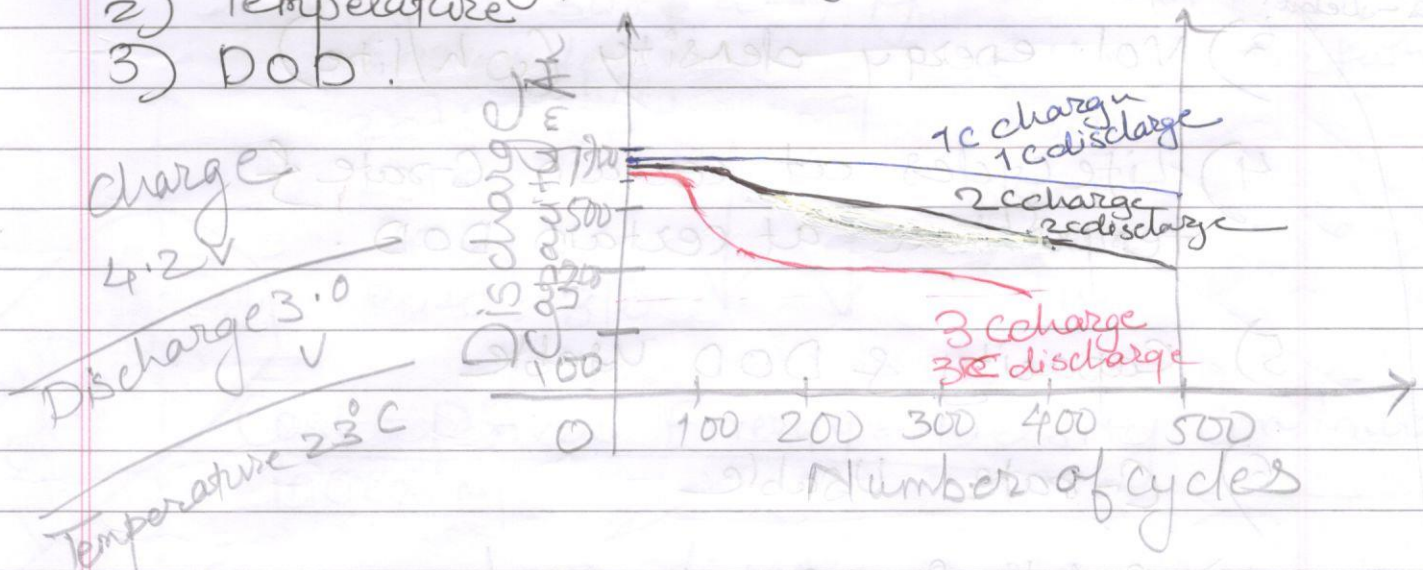


Battery cost

Battery capacity

Battery life : depends on

- 1) rate of charging/discharging
- 2) Temperature
- 3) DOD



Other key parameters :-

SoC

DOD

EOL

C-rate

cycle life

Aging - Calendar Life

Rechargeable batteries

- 1) NiCad
- 2) Nickel metal hydride
- 3) Lead Acid
- 4) Lithium Ion
- 6) Lithium Ion polymer
- 7) Solid state Lithium Ion

~~Battery~~

Parameters to select EV battery

1) Costs per kWh (Imp.)

2) Specific Energy density (Wh/kg)
gravitational energy density
 we carry with us → weight ↑ life ↑

3) Vol. energy density (Wh/litre)

4) Life cycles at certain C-rate & temperature at certain DOD.

5) Capacity & DOD usable

6) C-rate Usable

7) Safety & safety disposal.

→ Lithium, manganese, nickel etc. ^{phosphate, Cobalt} expensive
 that why weight of battery & also cost ↑.

If we need so but battery should be able to supply.

Battery Selected to match the vehicle requirement
 → in terms of energy required to travel certain range
 influencing battery size

→ In terms of instantaneous power requirements
 battery suddenly requires 100 kWh, (If we have 100 kWh)

Influencing C-rate

→ weight & size volume

→ Capital costs & Life cycles for battery replacement.

→ Time to recharge the battery

→ Safety & safe disposal

less than

1C would be great.

tea go
chademo
Japanese
as it's Japanese
Technology

Battery Specifications Price

Li-ion stands out on watt hour/kg & number of cycles

→ Touches 300 wh/kg.

Energy density
wh/kg

2011: 80

2015: 140

2018: 220

2020: 310

Cell price per kWh

\$ 800

\$ 275

\$ 140

\$ 110.

Battery
Cost \$ per year
pic.

Cost - Driver Energy - Density Continuous increasing.

Gravimetric ED of NMC & NCA cells is in betⁿ 250 to 300 wh/kg → moving to 400 - 500 wh/kg

(Putting Graphite, silica, anode)

Volumetric Energy density of NMC cells touching 500 wh/litre

~~Cost of battery inversely proportional to energy density.~~

~~→ Main driver cost reduction.~~

~~→ Higher energy density: Lower use of materials like Lithium, Cobalt, Nickel, Manganese, Graphite.~~

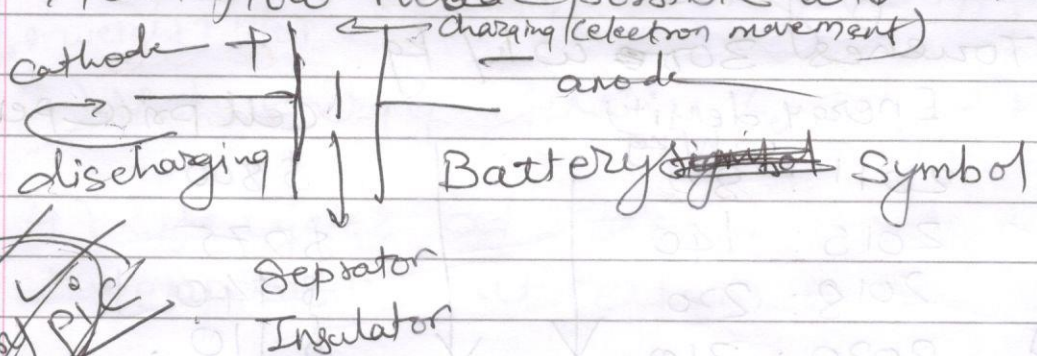
~~Comparison~~

Electrochemical Battery

Consists of Cathode, Anode & Electrolyte

→ Ions are atoms that have lost or gained electrons & thus electrically charged

: ION flow made possible with an electrolyte



~~Battery Chemistry Pic~~

~~Comparison of Li-ion batteries~~

The eight asks of a Battery.

High specific energy: - Long runtime (most are built batteries with high Ah)
(gravitational energy)

2. High specific power: - (For power tools & power provide high load capabilities) specific energy low. (higher discharging at maybe 4C etc)

3. Affordable price: - Materials, refining process manufacturing, quality control & cell matching add manufacturing cost. Volume production helps a bit. Single cell requires no cell match: lower cost.

4. Long Life. — High initial investment OK in countries with low interest rate. If battery life can be 20 years, low cost per year. Depends on battery design as well as temperature, charge times & harsh discharge rates.
5. Safety: → high specific energy systems are often reactive & unstable. ~~other~~
6. Wide operating range: — Cold temp — charging issue
High temp — shortens life
7. Low toxicity; — little toxic material but can be dangerous if disposed incorrectly
8. Fast charging; — only if batteries are build for it. (Aged & mismatched cells hurt during fast charging).

Battery
Comparison
& Cellulize

Batteries of future

- Need to compete with existing cells
In terms of specific energy, costs, number of cycles & temperature range.
- Nickel rich cathode } pushing up
→ Silica in anode. } energy density.

NMC111 to NMC433 to NMC532 to NMC632 to NMC8111 to Nickel rich cathode.

Li-Polymer (often Pouch cells) : Solid Electrolyte
(Gelly) - plastic like film.

- Poor conductivity at room temp (requires 60°C to enable cur
- Alternatively use gelled -
- Electrolyte gives slightly higher energy density (How)
thinner

Li-ion Battery cell construction.

Every cell has some container

Cylindrical/Pouch/prismatic

Cathode/Anode/seperator

Electrolyte betⁿ cathode & anode

LiPF₆ Lithium phosphorous
Fluoride.

Terminals -

Cylindrical - 18 650 ^{diameter mm} ^{length mm} (16 cm³ volume)
2.2/3.5 Ah, 3.7 volts. 10 to 11 Wh

Larger Battery 26 650

21700 defined by Tesla at 6 Ah
(24 cm³ volume)

Cylindrical cell has higher Energy-Density
than pouch & Prismatic

- But no. of cycles are usually less
- spot welding enough to make pack

Show pic of cells

- Pouch cells: soft, flatbody, 90 to 95%.
- Packing efficiency: Laser/Ultrasonic welded
 - : cells available at 24Ah, 44/50Ah & higher
 - : Used in larger vehicles
 - : Used to cost higher than cylindrical cells.

- Prismatic cells: Semi-hard plastic case with large threaded terminals.
- Cells from 7Ah, 15Ah, 24Ah, 30Ah, 40Ah, 44Ah, 50Ah
 - Used in 2 & 3 wheels
 - Can be spot welded, laser welded & ultrasonic welded.

- Li-Ion (NMC) Battery voltage
- 2.7V when SOC is 0% & 4.3V when SOC is 100%. ; nominal voltage 3.7V.
 - show inside cells pic

early
1900's

- Prismatic cell: — expensive, less efficient
thinner in design in thermal management.
- (swelling occurs)

show pic prismatic cell inside

Pouch cells: — show pic pouch cells

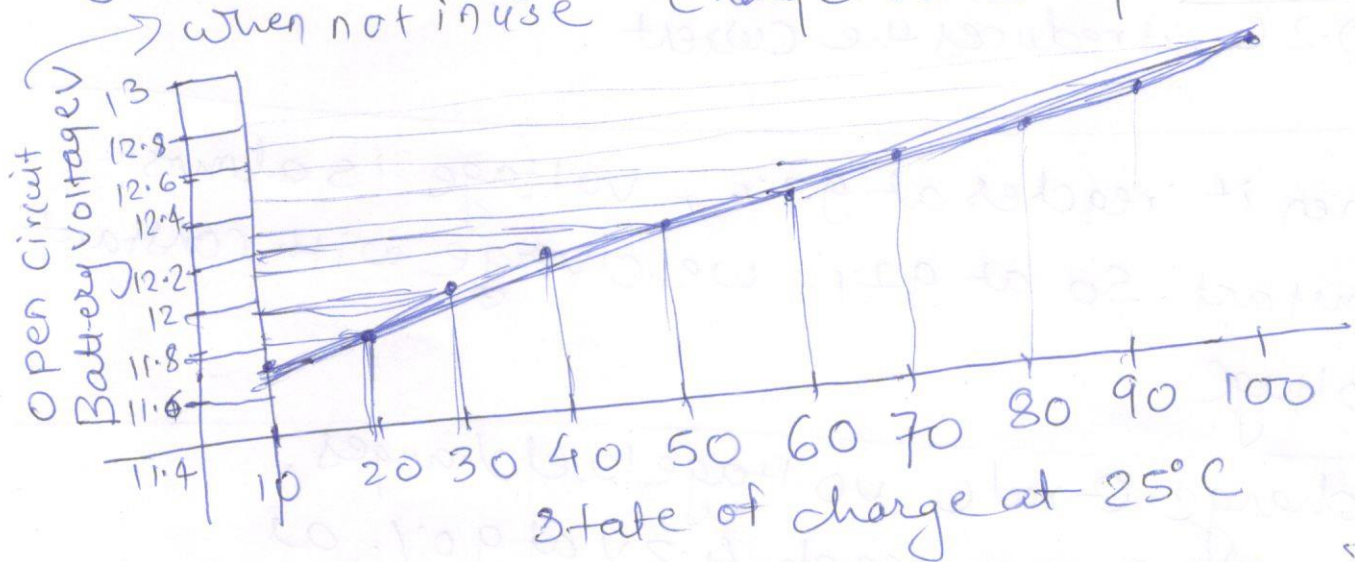
RC, failure of individual cells.

SOC } How to determine
 SoH } self discharge

SOC $\rightarrow$ percentage of total charge at present in the battery. (1)

What does 70% SOC implies?

e.g. laptop battery :- 11 mins. remaining, 10% charge remaining.



12V battery varies from 11.7V to 12.8V } Lead Acid Battery
 48V ————— 46.5V to 51.5V

$$V = OCV - IR \quad \left. \begin{array}{l} \text{Current} \\ \text{Internal resistance} \end{array} \right\} \text{when connected.}$$

V when connected = Open circuit voltage = (internal resist $\times$ Current)

This linearity is not there in Li-Ion
 In Li-Ion, BMS will measure OCV of battery, convert it in number & transmit through CAN

$\rightarrow$ LI-Ion voltage vs SOC Pic show
 when charging is done at slow rate, negligible current, then OCV = voltage of battery. (zero)

(2)

A slow charging at 5%. So C you are already at 3.5V, After that it $\uparrow$ linearly at linear line $\rightarrow$ constant current charging.

Initially it tries to adjust the current to give $0.2C$. It increases to reach $0.2C$, when exceeds $0.2C$ it reduces the current.

When it reaches at 97%, voltage is almost constant. So at 97% we charge with constant voltage.

If we change C-rate voltage level changes. e.g. at 0.5C you reach 4.2V at 90% as oppose to 97% in 0.2C

While discharging: - at 0.2C discharging, at 10% SOC, voltage is nearly 3.4V
at 2.3C discharge it will be 2.9V at 10%.

SOC has to be measured at open circuit voltage

- Constant Current (CC) charging at high rate (assumed $2C$)
- $\rightarrow$ only partially charge battery: possible only up to some low SOC (57%)
 - $\rightarrow$ Beyond that it will be a Constant Voltage (CV) charging, which is very low current charging.
 - $\rightarrow$ High rate charging is only meaningful for large batteries

High-rate discharge also hurts battery.

Show pic discharge capacity vs no. of cycles

Energy pumped into cell betⁿ 3.5V & 4.2V when slow charged

→ for fast charge, it is betⁿ 3.9V & 4.2V

→ slow discharge energy is betⁿ 3.4V & 4.1V

SOC is not linear fⁿ of voltage

How one can measure SOC?

→ ~~constant~~ voltage method

→ obtain open circuit cell voltage (OCV) Vs SOC accurately in lab at very low charging rate (C/25 going to C/100) for different temperatures.

* Does OCV vs SOC depend on SOH

: not clear - conflicting opinions among researchers

→ SOC is nonlinear fⁿ of OCV, only when battery is fully at rest (very slow charge or discharge is ok)

Coulomb counting

Coulomb

Very accurate but dependent on accurate SOH & precision of current measurement.

→ Measuring current (total coulombs) flowing in & out of battery: gives one a change in SOC if SOH as well as the initial capacity is known

reference is assumed

(start at one charge level & then measure SOC at some time)

In lab - measure initial SOC ④
~~for~~ making sure you measure OCV with
no charge & from there you can
start counting coulombs coming in/out.

So knowing correct starting point is
important. (so start at 100% charged as
initial SOC)

coulomb count betⁿ 2 instants of SOC will be
give amount of energy removed or added
from battery.

Change in SOC (or ΔSOC) = $\frac{\text{charge pumped in or out of battery}}{\text{Capacity} \times \text{SOH}}$ ^{initial} $\rightarrow$ it's new tool.
_{if not new find out eq. 0.9}

$\rightarrow$ Need to be converted to %.

$\rightarrow$ charge pumped in or out is ~~coulomb count~~
coulomb count * electron charge (or integration of current over time)

computed charge is ~~IN~~, ~~the~~ ΔSOC is +ve,

else it ~~is~~ -ve.

$\text{SOC}_{\text{new}} = \text{SOC}_{\text{old}} + \Delta \text{SOC}$ (need to know SOH
to correctly know
 ΔSOC is dependent
on SOH)

Repeated charging & discharging will reduce accuracy as error builds up.

(5)

→ error in current measurement & degradation in SOH
Creates & keeps on creating error.

Solution → take battery again to 100 or zero & start again.

-
- 1) nonlinear relation betn SOC & V
 - 2) we know relation betn SOC & V only when current is zero.
 - 3) with charging & discharging SOH goes down & its imp we know it accurately.

see Li-ion charging graph

Check at 4V with different charging rate
SOC is different.

-
- 4) initial SOC has to be known.

SOH? How to determine it.

Not accurately but we can!

It is age/health of a battery told by "SOH"
"it represents amount ^{by which} battery has deteriorated due to irreversible physical or chemical change".

periodically completely discharge & then slowly charge the battery (track OCV) & then discharge slowly & carry out Coulomb count.

- Give several hours rest after full charge
- Indicates max. charge that battery can hold

Alternate Method internal resistance (6)

- As battery electrodes deteriorate, its Capacity to deliver current ~~also~~ also reduces
- Internal resis. of cell indicates the capability to deliver current.
- Difference betⁿ internal resis. of fresh & used cell, helps in estimating SOH
show impedance change vs battery life p.c.

Battery pack development

(7)

pic → what makes a battery pack

Safety issues

Cell balancing → work for small difference
→ Not for large difference.

Careful electrical design so that every cells gets equally charged / discharged.

cell balancing :- small change in internal resis. etc creates imbalance.

→ when pack temp ↑, cell temp ↑, cells behave good only when at 25°C. So this design is crucial.

→ When we run vehicle, the vehicle gets vibration. which makes pack & hence cells vibrate.

→ if the cells bulge or swell & vibrate it will be disastrous.

→ 100 Rs battery → 35 Rs for battery pack design.
25 Rs for making cells
40 Rs for material.

Thermal design :- Remove heat from pack immediately
cells → need temp control.

Mechanical design :- Right pressure need to be applied to cells, else they will bulge

BMS :- 1) only balanced cells used in packs

Require :- voltage / current / temperature monitoring of each cell & balancing & cells during charging as well as discharging

- pack should cut off if the temp. ↑
- Communicates with charger to decide charging Strategy.

Battery pack design Electrical

certain voltage & capacity
48V battery, 5kW battery, 100A is current.

Battery capacity is 5kW

at times you might need to push 8kW or 9kW
it will be nearly 200A

with this high current comes big problems
~~even the~~

→ high current implies high ~~impt~~ ohmic losses (Thick cable) keep it < 200 Amps

→ $3\text{m}\Omega \times 200\text{A}$
 $= 0.6\text{V}$ at cell } let's say there is a problem in connecting the busbar there will be I^2R losses also.

48V to 72V → for 2W or 3W
small 4W

100Wh to 15kWh: 1kWh requires 21Ah cells
possible with cylindrical & prismatic

350 V for medium sized batteries

(9)

for larger cars & pickups

20kwh to 70kwh batteries:

1kwh battery requires $< 3Ah$

750V for (M) for buses & trucks

60kwh battery size to 300kwh.

two many cells if cylindrical used

Build Battery packs from cells

Cell volt. = $3.7V$ ($3.1V$ to $4.1V$)

cell capacity $3.4Ah$ to $50Ah$ (prismatic/pouch)

→ Requires cells connected in series to get

high voltage

14 cells in series is $51.8V$ [so $3.7V$ in series will require 100 cells to get $370V$]

→ Requires cells in parallel for higher capacity

8 cells (^{of this capacity} $50Ah$) in parallel gives $400Ah$.

If you want $400Ah$ for $3.4Ah$ cylinder cell then you need 125 cells in parallel.

$\frac{m p n s}{n s m p}$
 first m cells in parallel } form module 10
 & then put this module in series
 n cells in series to form strings
 & then connect strings in parallel.

4P14S pack with 3.7 V cylindrical of 3.4 Ah
 gives

Capacity of Ah is $4 \times 3.4 \text{ Ah}$ or 13.6 Ah

then voltage $14 \times 3.7 \text{ V}$ or 51.8 V

Total capacity = $13.6 \text{ Ah} \times 51.8 \text{ V}$
 = 704.5 Wh

Same in capacity for 14S4P same cells.

Building $m s n p$ battery pack

Cells connected in series to form STRING

14 cells connected in series to form 48 V Battery

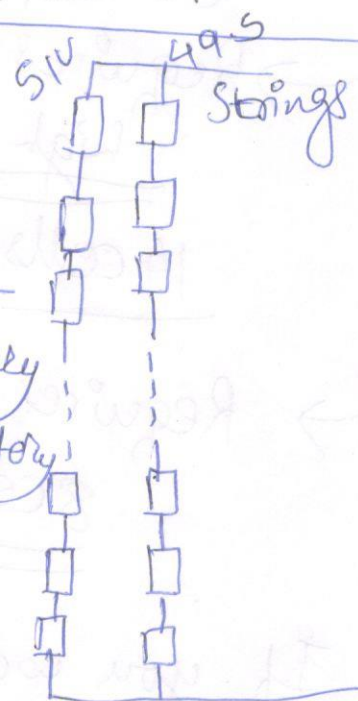
20 cells

72 V Battery

$14 \times 3.7 = 51.8$ close to 48 V

$20 \times 3.7 = 74 \text{ V}$ close to 72 V

Find 100 cells & 200 cells in series



2, 14S2P strings with 15 Ah cells would be $48 \text{ V} \times 2 \times 15$
 = 1.5 kWh
 capacity

Concern

51V & 49.5V

(11)

If strings do not have exactly same voltage
current will flow from one string to
another for balancing → will happen continuously while charging or discharging even when IDLE
Not good for battery

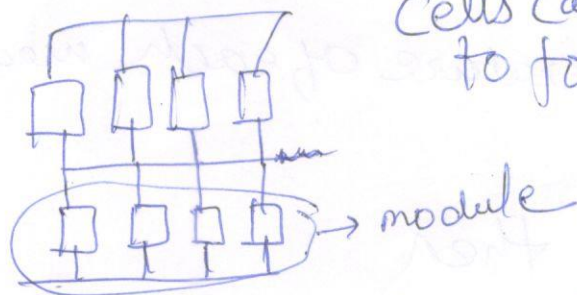
Solution :- Take all cells of same voltage
BUT

over the period after some use again
imbalance among cells can occur / occurs

Better option

NPMS

Cells can be connected in parallel
to form modules.



Four 15Ah cells gives module 60Ah

Eight 15Ah ———— 120Ah

Sixteen 15Ah ———— 240Ah

Modules now can be connected in series to make battery of higher voltages

(12)

→ If we connect 14 modules of 60Ah (4 cells of 15Ah in parallel) in series, we get $15V \times 60Ah = 3.06 kWh$

→ If we connect 14 modules of 120Ah in series $51V \times 120Ah = 6.12 kWh$.

Not major balancing issue here

of course autobalancing betⁿ cells in a module in cells

~~to~~ detection

* Multiple cells packed in 11el to form a module.
→ cells selected so that they are of same voltage
→ Cells connected with metal bar that conducts electricity.

When during charging unequal current flows,
(betⁿ multiple modules)

we do equalization

→ monitor voltages & temperature of each module & total pack current

If a module is overcharged then

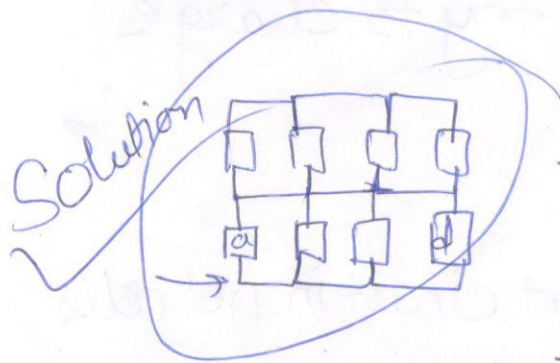
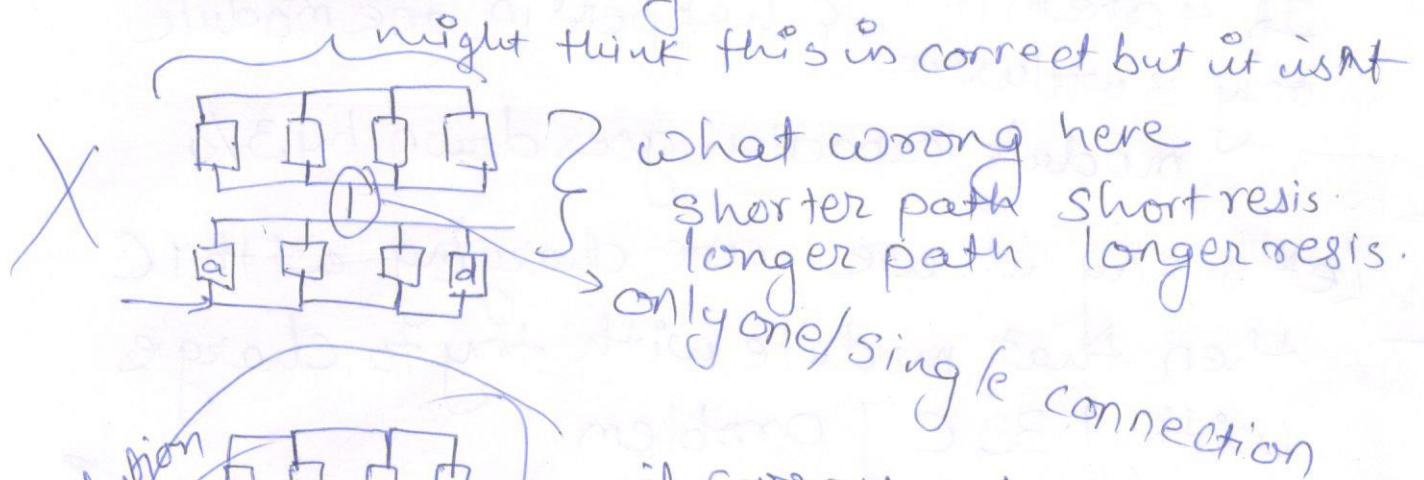
passive balancing

Active balancing (DC-DC converter)

BMS limit temperature of each module if active

cooling is done

- High currents are involved.
- conductor is not zero voltage drop
- smallest resistance difference betⁿ 2 electrical paths may result into differential voltage drops
- Currents will flow more on one rather than the other creating imbalances.



→ Electrical path to cell marked 'a' is shorter than that for one marked 'd'

→ Cell imbalance will be created betⁿ 'a' & 'd'

It happens more betⁿ modules, as current from one cell in a module to other is more as compared to that from another cell.

Continuous imbalance will quickly deteriorate capacity

Failures in battery pack:

(14)

1) If a cell in $nPmS$ pack

→ one of n parallel cells maybe open circuit

★ $(n-1)$ cells in module continue to function

★ module capacity goes down by $(1/n)Ah$

★ Battery will fⁿ with $(n-1)PmS$

If 4 are in 11st } happens in one module
only 3 will work }
module capacity goes down by $3/4$

[Let's say if we are charging with 1C
then that module will try to charge
with 1.33 C] problem

2) If a cell in $nPmS$ pack

→ one cell becomes short circuit in 11st cells

★ Full module has zero voltage

★ Can continue to fⁿ

★ 'only if' the drive train fⁿs at low volt.

[it is a poor performance]

Battery voltage will be $(m-a)S$

& the pack is $nP(m-1)S$

yet better
than OC

3) Repair :- replace whole module
cannot replace 'cell'

~~While~~ While designing Battery you have to
take in to account what if I need to
replace something.

If a serial connection betⁿ module fails :-
battery fails

4) BMS Fails!!! [replace BMS]

MOSFET in BMS heated; design issue
Temperature Sensor failure

5) Cell capacity deteriorate

6) In correct SOC & SOH estimation
